HDOC Day-67 Assignment

Batches 15 & 16 | C Programming - Recursion

Question

Write a recursive function to find the GCD of two positive integers using Euclid's algorithm.

Aim

To calculate the Greatest Common Divisor (GCD) of two positive integers using a recursive function in C.

Algorithm

1. Read two positive integers a and b.
2. If b is 0, return a.
3. Otherwise, call gcd(b, a % b) recursively.
4. Print the returned GCD.

C Program

```
#include <stdio.h>

int gcd(int a, int b) {
    if (b == 0) {
        return a;
    }
    return gcd(b, a % b);
}

int main() {
    int a, b;
    printf("Enter two positive integers: ");
    scanf("%d %d", &a, &b);

    if (a <= 0 || b <= 0) {
        printf("Please enter positive integers.\n");
        return 1;
    }

    printf("GCD of %d and %d is %d\n", a, b, gcd(a, b));
    return 0;
}
```

Sample Output

```
Enter two positive integers: 48 18
GCD of 48 and 18 is 6
```

Working

gcd(48, 18) = gcd(18, 12) = gcd(12, 6) = gcd(6, 0) = 6. The base case b == 0 stops recursion and returns the GCD.

Conclusion

The program successfully finds the GCD of two positive integers using Euclid's recursive algorithm.